

## Task – 5.2c

Name – Nirosh Ravindran

1. It provides a secure gateway for accessing a private network from an external network. It mitigates unauthorized access and potential attacks on resources within the private network.

In case if we want to use a Linux instance in a private network via SSH from an external network, we can use bastion host to control and secure these connections.

Bastion hosts mainly run on public subnet on an EC2 instance. When we want to access the Linux instance, we should setup the instance in a way where the SSH traffic is first directed towards the bastion host first and then reaches the Linux instance. This will add an extra layer of security to the

2. It provides a static and public IPv4 address that is designed for dynamic cloud computing. It associates the address with any instance or network interface in our account's VPC.

An instance's address can be switched quickly if one fails. We will be able to move all attributes of the network interface from one instance to another in a single step when we associate the elastic IP with the network interface instead of directly with an instance.

This helps in managing instances and maintaining uninterrupted service.

3. SSH pass through is needed to securely access resources within a private network or subnet from an external network or a local computer. Using this, we establish a secure connection to a public host like bastion host to reach the private resources.

When data is being in transit over an untrusted network, SSH will ensure data remains protected by encrypting the communication between the host and target. This will also provide a controlled access point, reducing the attack surface and enhancing overall security posture.

# Module 6 - Challenge Lab: Creating a VPC Networking Environment for the Café

## Task 1: Creating a public subnet

We're setting up a public subnet in our Lab VPC to connect to the internet. First, we create the subnet with specific settings like its name and IP address range.

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VPC dashboard

EC2 Global View

Filter by VPC: Select a VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

Your VPCs (1/2) Info

Search

|                                     | Name    | VPC ID                | State     | IPv4 CIDR     | IPv6 CIDR | DHCP opti |
|-------------------------------------|---------|-----------------------|-----------|---------------|-----------|-----------|
| <input type="checkbox"/>            | -       | vpc-0368c4494a6c72f01 | Available | 172.31.0.0/16 | -         | dopt-00a4 |
| <input checked="" type="checkbox"/> | Lab VPC | vpc-06142578ccb2b51ee | Available | 10.0.0.0/16   | -         | dopt-00a4 |

vpc-06142578ccb2b51ee / Lab VPC

DetailsResource mapCIDRsFlow logsTagsIntegrations

Details

|                               |                                     |                       |                                  |
|-------------------------------|-------------------------------------|-----------------------|----------------------------------|
| VPC ID                        | State                               | DNS hostnames         | DNS resolution                   |
| vpc-06142578ccb2b51ee         | Available                           | Enabled               | Enabled                          |
| Tenancy                       | DHCP option set                     | Main route table      | Main network ACL                 |
| Default                       | dopt-00a4556577a75ed09              | rtb-067b7509d1f77b616 | acl-056b1f206dd295756            |
| Default VPC                   | IPv4 CIDR                           | IPv6 pool             | IPv6 CIDR (Network border group) |
| No                            | 10.0.0.0/16                         | -                     | -                                |
| Network Address Usage metrics | Route 53 Resolver DNS Firewall rule | Owner ID              |                                  |

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You have successfully created 1 subnet: subnet-0cfa9ab65be3ae8b3

Subnets (1) Info

Find resources by attribute or tag

Subnet ID : subnet-0cfa9ab65be3ae8b3

Clear filters

|                          | Name          | Subnet ID                | State     | VPC                             | IPv4 CIDR   |
|--------------------------|---------------|--------------------------|-----------|---------------------------------|-------------|
| <input type="checkbox"/> | Public Subnet | subnet-0cfa9ab65be3ae8b3 | Available | vpc-06142578ccb2b51ee   Lab ... | 10.0.0.0/24 |

Select a subnet

CloudShell

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Then, we add an internet gateway to our VPC to enable internet access for the subnet. Finally, we update the route table to direct internet-bound traffic through the gateway.

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Select a VPC

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Egress-only internet gateways

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Peering connections

Internet gateway igw-06ebf256c49c11de2 successfully attached to vpc-06142578ccb2b51ee

VPC > Internet gateways > igw-06ebf256c49c11de2

igw-06ebf256c49c11de2 / IGW

Details info

Internet gateway ID  
igw-06ebf256c49c11de2

State  
Attached

VPC ID  
vpc-06142578ccb2b51ee | Lab VPC

Owner  
584532640012

Tags

Search tags

Manage tags

< 1 > ⚙

| Key  | Value |
|------|-------|
| Name | IGW   |

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Filter by VPC:  
Select a VPC

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Peering connections

Updated routes for rtb-067b7509d1f77b616 / Public route table successfully

VPC > Route tables > rtb-067b7509d1f77b616

rtb-067b7509d1f77b616 / Public route table

Details info

Route table ID  
rtb-067b7509d1f77b616

Main  
Yes

Explicit subnet associations  
-

Edge associations  
-

VPC  
vpc-06142578ccb2b51ee | Lab VPC

Owner ID  
584532640012

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Both Edit routes

< 1 > ⚙

| Destination | Target                | Status | Propagated |
|-------------|-----------------------|--------|------------|
| 0.0.0.0/0   | igw-06ebf256c49c11de2 | Active | No         |
| 10.0.0.0/16 | local                 | Active | No         |

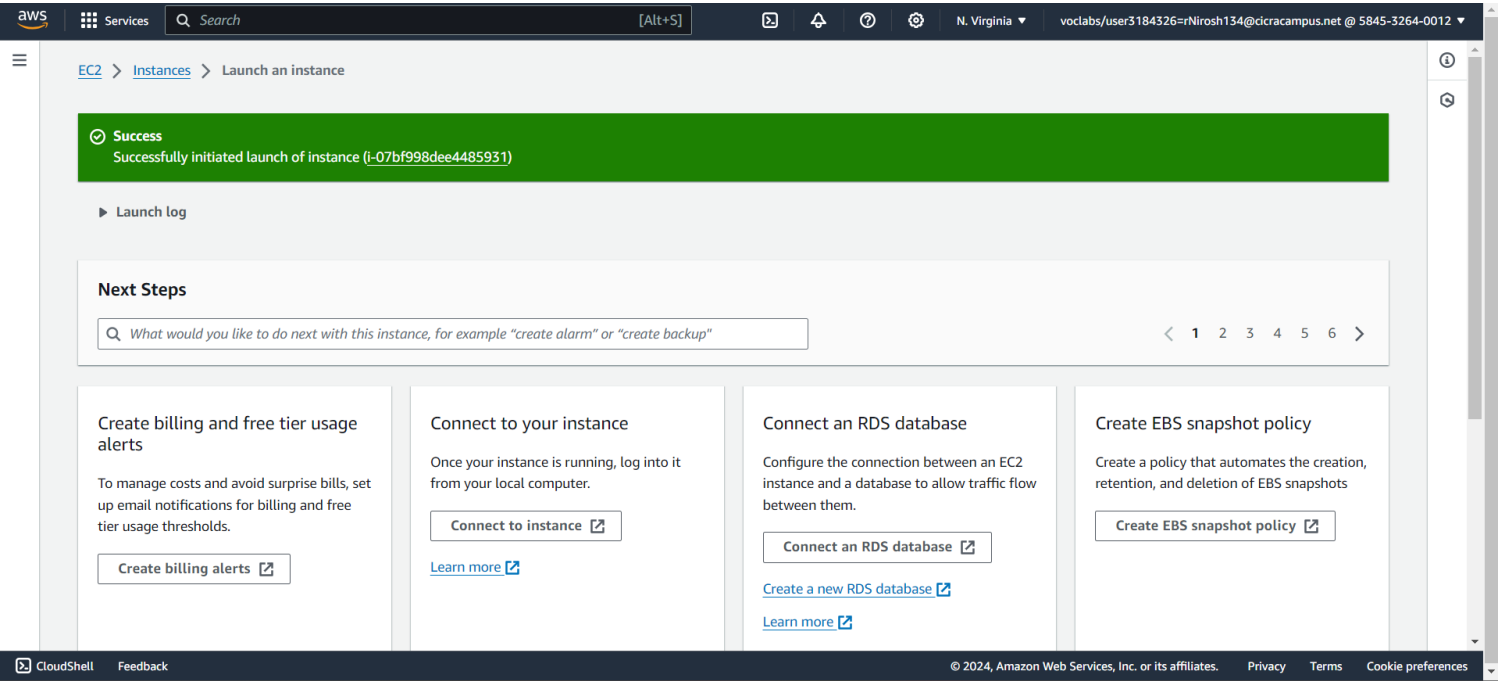
CloudShell

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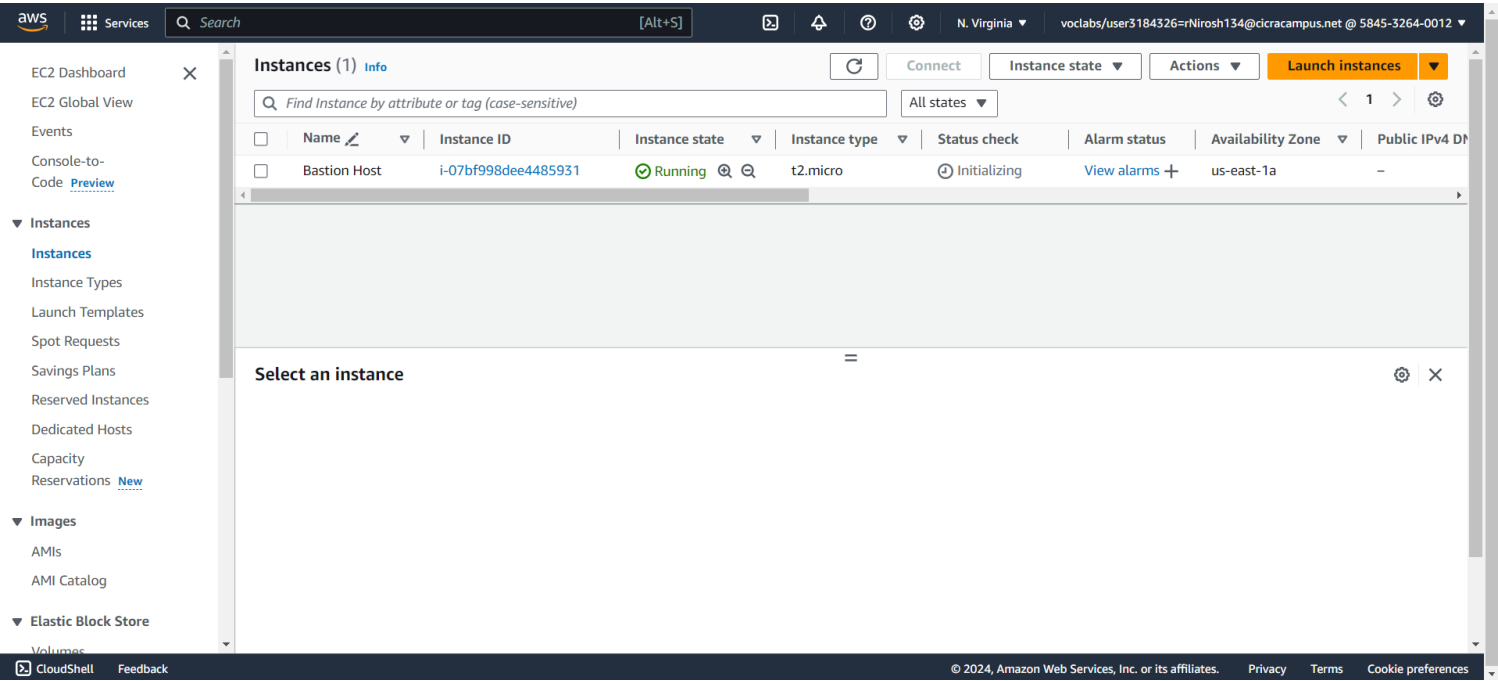
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Task 2: Creating a bastion host

We're creating a bastion host in the Public Subnet of our Lab VPC using Amazon EC2. This host acts as a secure gateway for accessing instances in private subnets.

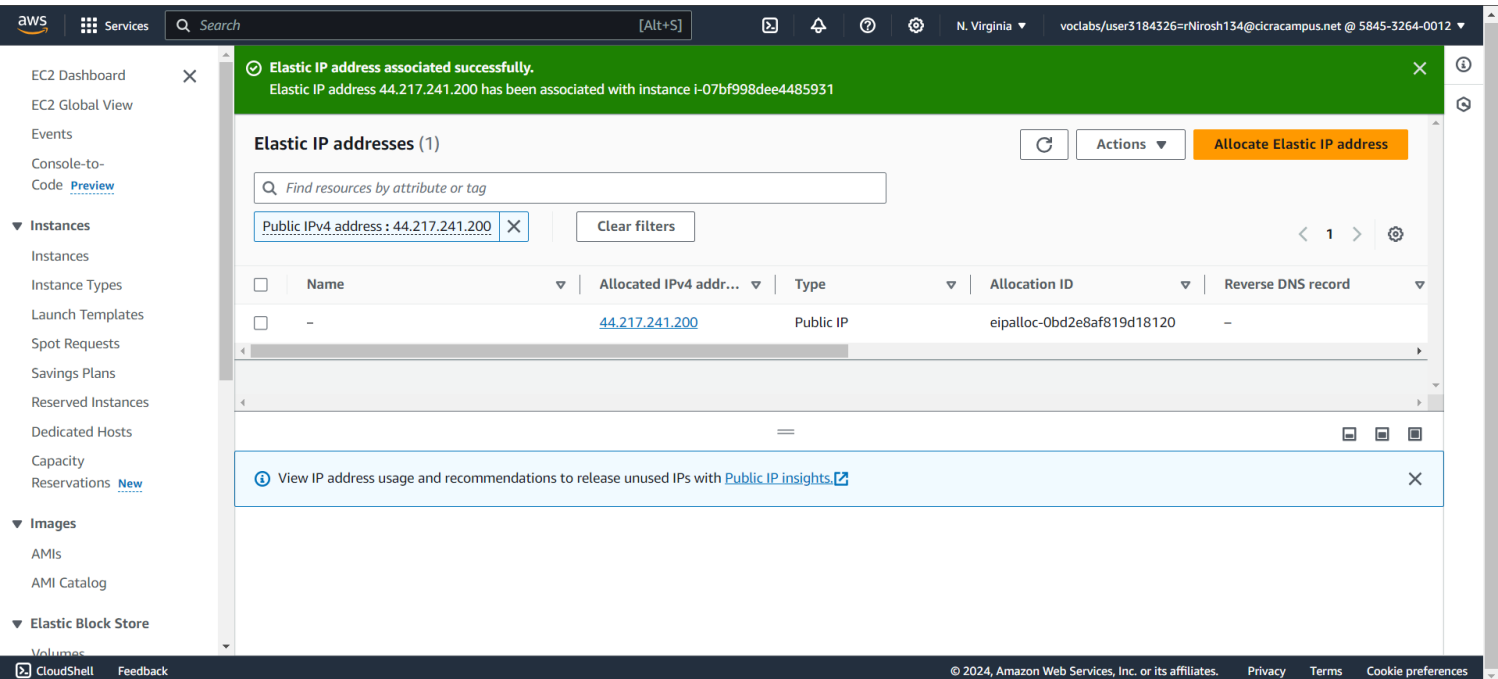
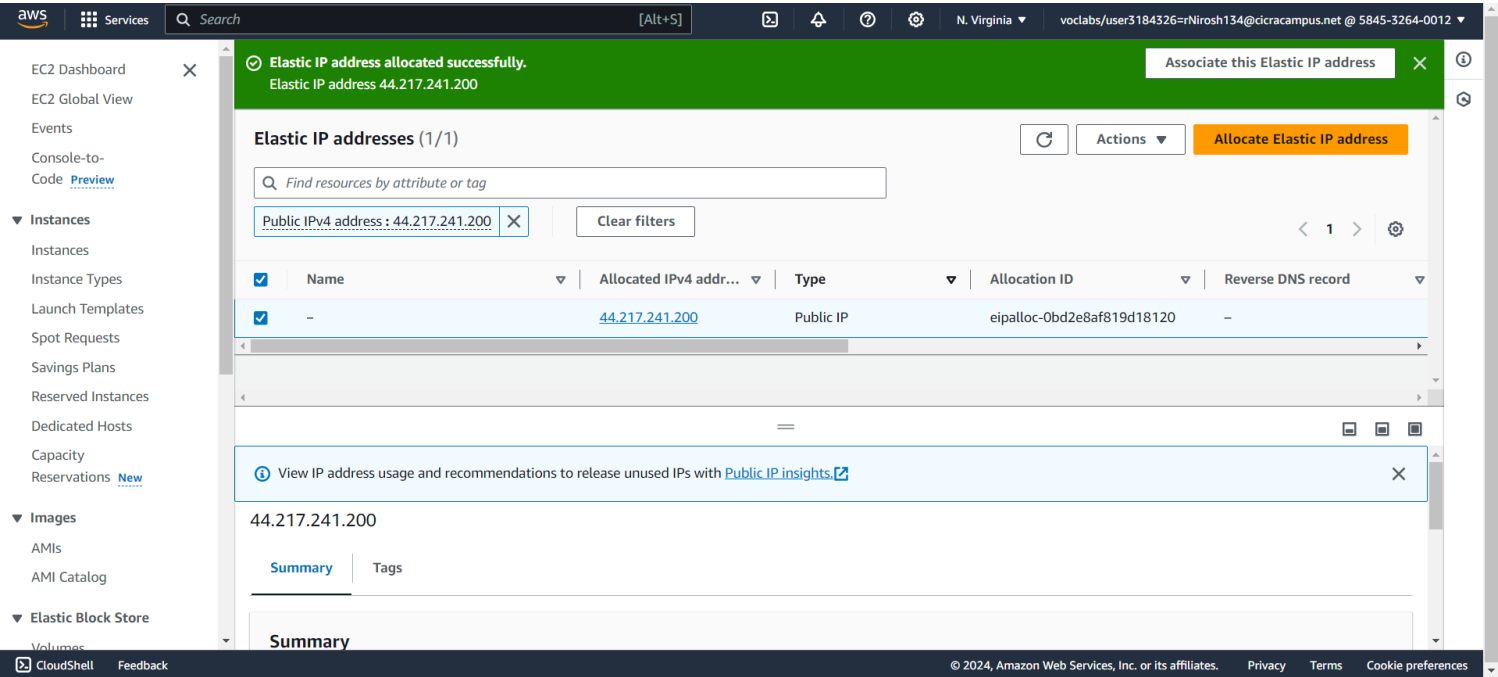


We then choose specific settings for the instance, like its type and security group, to ensure only authorized access.



### Task 3: Allocating an Elastic IP address for the bastion host

In this task, we're giving our bastion host a special kind of address called an Elastic IP (EIP). Right now, the bastion host can't be accessed from the internet because it doesn't have a public address.



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EC2 Dashboard

EC2 Global View

Events

Console-to-Code [Preview](#)

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity

Reservations [New](#)

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

CloudShell

Feedback

Instances (1/1) [Info](#)

Refresh

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

Instance state = running

Clear filters

<

1

>

| <input checked="" type="checkbox"/> | Name         | Instance ID         | Instance state | Instance type | Status check      | Alarm status                | Availability Zone | Public IPv4 D      |
|-------------------------------------|--------------|---------------------|----------------|---------------|-------------------|-----------------------------|-------------------|--------------------|
| <input checked="" type="checkbox"/> | Bastion Host | i-07bf998dee4485931 | Running        | t2.micro      | 2/2 checks passed | <a href="#">View alarms</a> | us-east-1a        | ec2-44-217-241-200 |

i-07bf998dee4485931 (Bastion Host)

Details

Status and alarms [New](#)

Monitoring

Security

Networking

Storage

Tags

Instance summary [Info](#)

Instance ID

i-07bf998dee4485931 (Bastion Host)

IPv6 address

-

Hostname type

IP name: ip-10-0-0-107.ec2.internal

Answer private resource DNS name

Public IPv4 address

44.217.241.200 [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-0-107.ec2.internal

Instance type

Private IPv4 addresses

10.0.0.107

Public IPv4 DNS

ec2-44-217-241-200.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses

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## Task 4: Testing the connection to the bastion host

In this task, we'll use a special file (SSH key) to test our connection to the bastion host. We should download the PPK file from the instructions.

After that, this file will be used to connect to your bastion host using SSH. Then we will test the connection using the putty terminal

[illegible]

## Task 5: Creating a private subnet

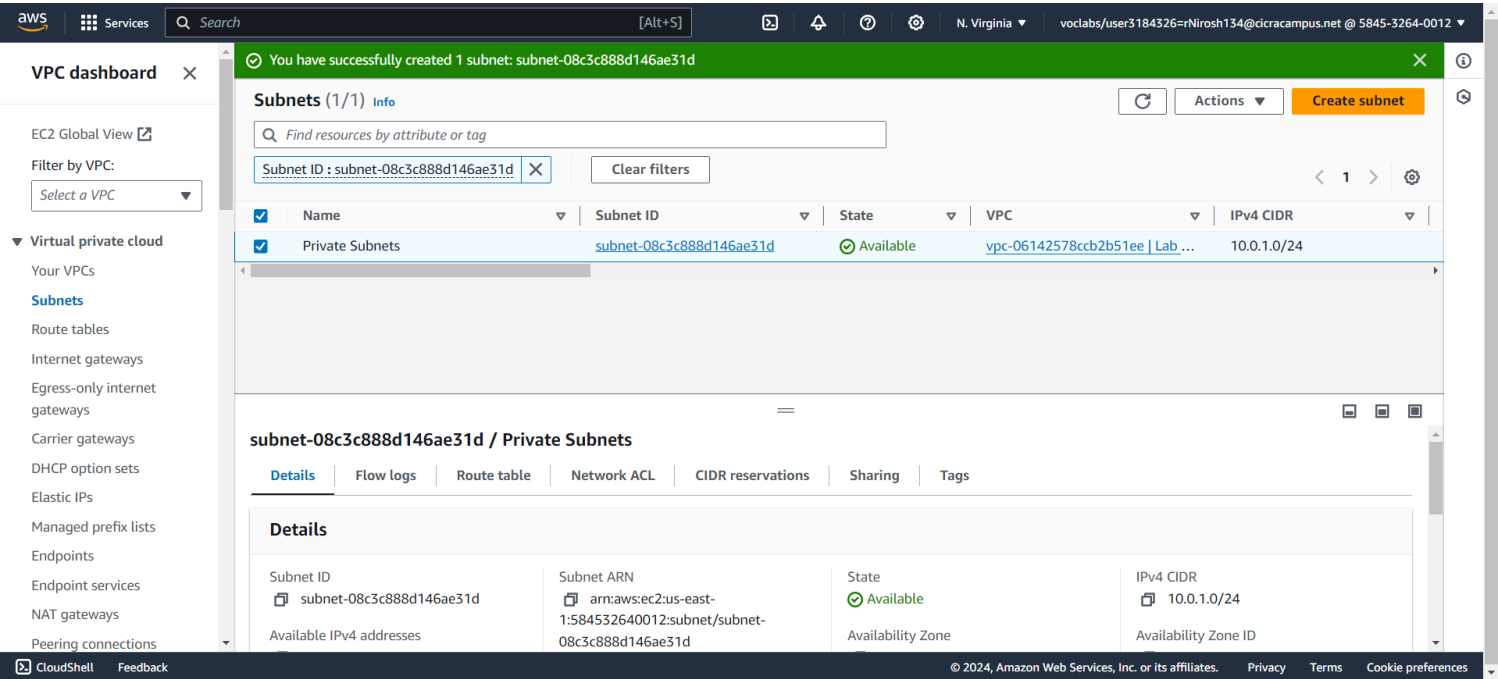
In this task, we're making a private subnet within our Lab VPC.

To do this, we create a private subnet in the console with specific settings:

We name it 'Private Subnet'.

It's in the same availability zone as the public subnet.

The IP address range for this subnet is 10.0.1.0/24.

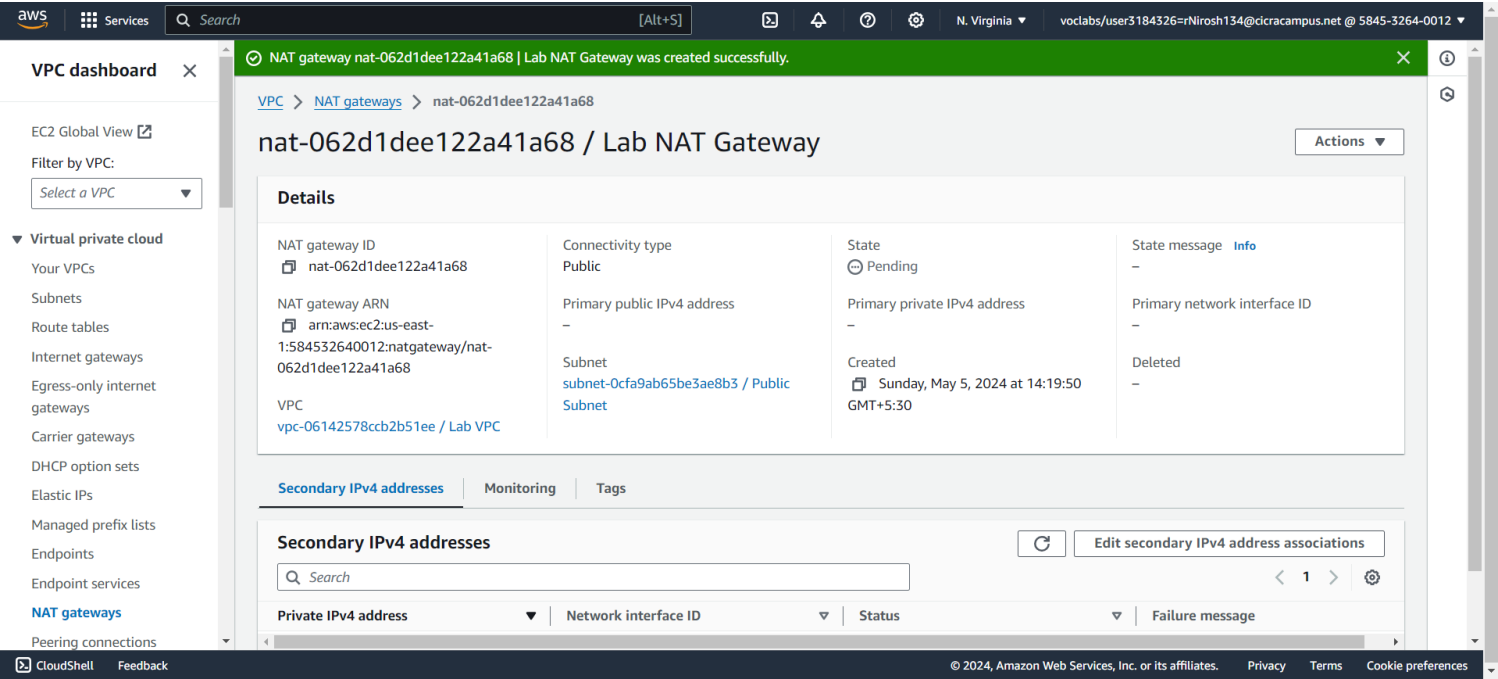




Task 6: Creating a NAT gateway

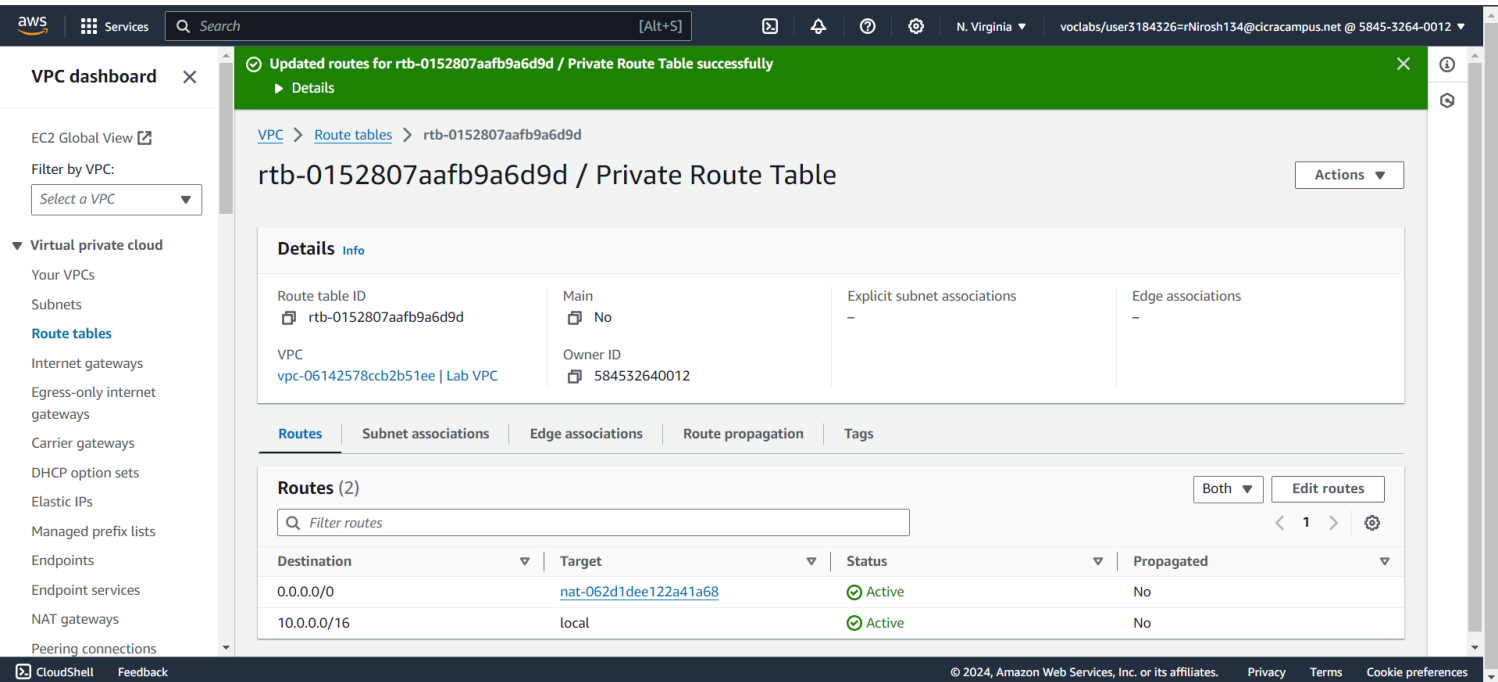
In this task, we're setting up a way for resources in our Private Subnet to access the internet.

First, we create a NAT gateway named 'Lab NAT Gateway' in our Public Subnet. This gateway needs an Elastic IP address.



Next, we create a new route table named 'Private Route Table' that directs all internet-bound traffic (0.0.0.0/0) to the NAT gateway.

Finally, we attach this route table to the Private Subnet we created earlier. This ensures that resources in the Private Subnet can reach the internet through the NAT gateway.

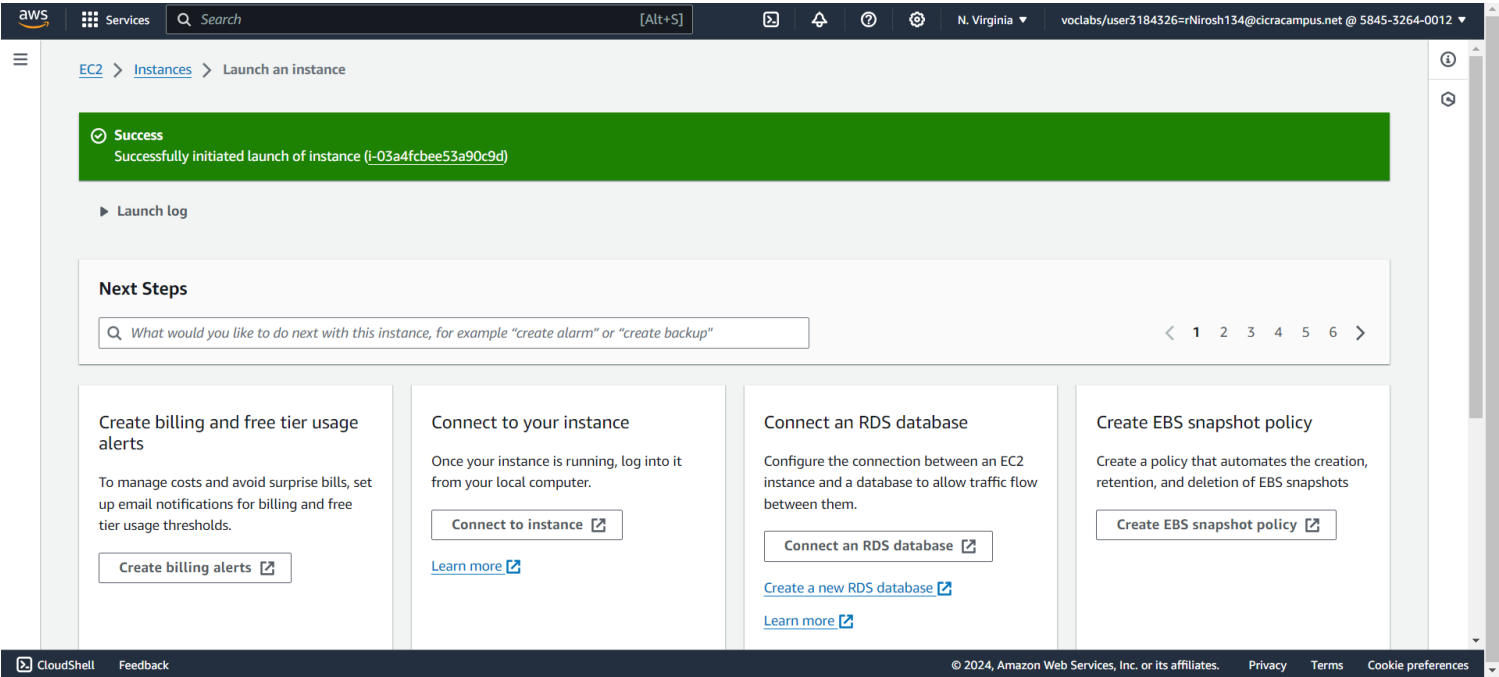


Task 7: Creating an EC2 instance in the private subnet

In this task, we're setting up a new EC2 instance in our Private Subnet. We'll configure it to allow SSH traffic from the bastion host. Also, we'll create a new key pair to access this instance.

First, we create a new key pair named 'vockey2' and download the appropriate file for our operating system.

At last, we create the EC2 instance with the specified configurations



## Task 8: Configuring your SSH client for SSH passthrough

To access the private instance securely without uploading the key pair to the bastion host, we'll use SSH passthrough. This allows us to use a key pair stored on our computer.

for that we use a software called Pageant from the PuTTY.

```
ec2-user@ip-10-0-0-107:~  
login as: ec2-user  
Authenticating with public key "imported-openssh-key" from agent  
#  
~\####_ Amazon Linux 2023  
~~\#####\  
~~\####|  
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~'-'>  
~~~  
~~-.-  
~~/ /  
~~/m/'- /  
Last login: Sun May 5 08:43:58 2024 from 223.224.27.250  
[ec2-user@ip-10-0-0-107 ~]$
```

## Task 9: Testing the SSH connection from the bastion host

In this task, we're testing the SSH connection from our bastion host to the EC2 instance in the Private Subnet.

First, we connect to the bastion host using SSH, using the method described in the SSH passthrough section.

Then, we connect to the private instance using SSH and its private IP address.

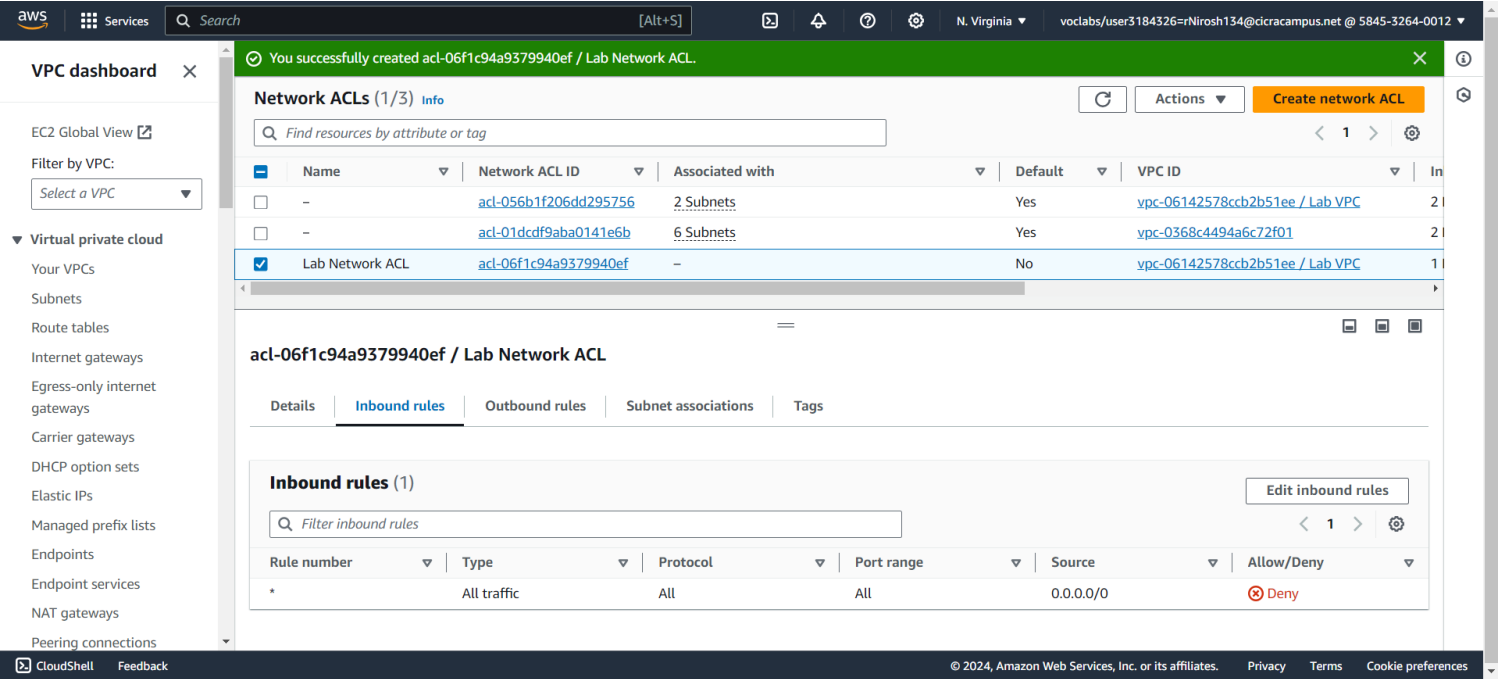
Once connected to the private instance, we test its internet connection by pinging 8.8.8.8.

This establishes communication between the Bastion Host in the Public Subnet and the EC2 instance in the Private Subnet

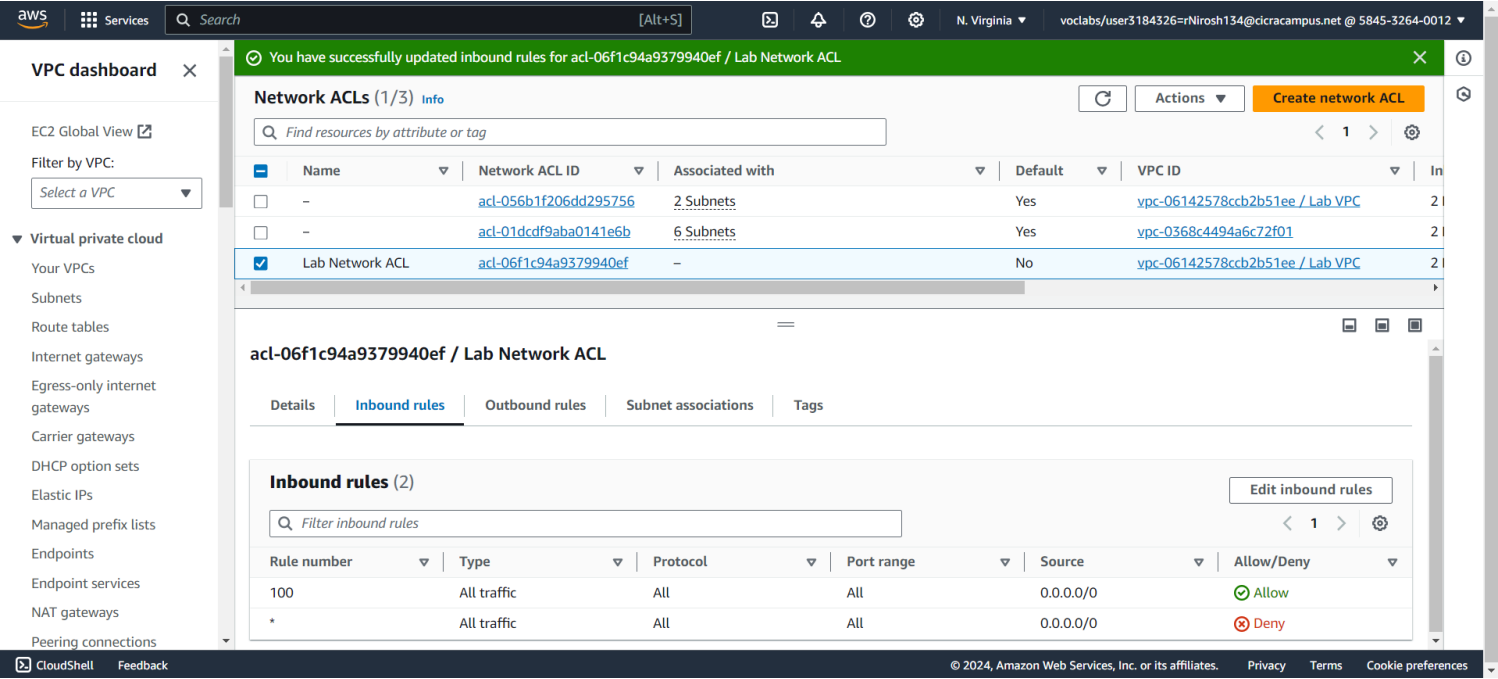
```
ec2-user@ip-10-0-1-160:~  
The authenticity of host '10.0.1.160 (10.0.1.160)' can't be established.  
ED25519 key fingerprint is SHA256:tAtBrAbOmT2CIcjoWiwZ3zxHMittEbkL848So8WV0lc.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.1.160' (ED25519) to the list of known hosts.  
  
#  
~\#### Amazon Linux 2023  
~~\#####  
~~\###|  
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~'-'->  
~~~  
~~.-./-/  
~/m/'-/  
[ec2-user@ip-10-0-1-160 ~]$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=3.00 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=1.76 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=1.69 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=1.70 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=1.73 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=1.73 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=115 time=1.75 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=115 time=1.83 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=115 time=1.78 ms  
^C  
--- 8.8.8.8 ping statistics ---  
9 packets transmitted, 9 received, 0% packet loss, time 8014ms  
rtt min/avg/max/mdev = 1.687/1.884/2.995/0.394 ms  
[ec2-user@ip-10-0-1-160 ~]$
```

Task 10: Creating a network ACL

First, we'll create an EC2 instance in the Public Subnet and configure a security group to allow ICMP traffic from the local network.



Then, we'll create a custom network ACL called 'Lab Network ACL' for the Lab VPC. By default, this ACL will deny all traffic.



Finally, we'll configure the custom network ACL to allow all traffic into and out of the Private Subnet.

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EC2 Global View

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Peering connections

You have successfully updated outbound rules for acl-06f1c94a9379940ef / Lab Network ACL

Network ACLs (1/3)

Find resources by attribute or tag

1

|                                     | Name            | Network ACL ID        | Associated with | Default | VPC ID                          | In |
|-------------------------------------|-----------------|-----------------------|-----------------|---------|---------------------------------|----|
| <input type="checkbox"/>            | -               | acl-056b1f206dd295756 | 2 Subnets       | Yes     | vpc-06142578ccb2b51ee / Lab VPC | 2  |
| <input type="checkbox"/>            | -               | acl-01dcdf9aba0141e6b | 6 Subnets       | Yes     | vpc-0368c4494a6c72f01           | 2  |
| <input checked="" type="checkbox"/> | Lab Network ACL | acl-06f1c94a9379940ef | -               | No      | vpc-06142578ccb2b51ee / Lab VPC | 2  |

acl-06f1c94a9379940ef / Lab Network ACL

Details | Inbound rules | Outbound rules | Subnet associations | Tags

Outbound rules (2)

Edit outbound rules

Filter outbound rules

1

| Rule number | Type        | Protocol | Port range | Destination | Allow/Deny |
|-------------|-------------|----------|------------|-------------|------------|
| 100         | All traffic | All      | All        | 0.0.0.0/0   | Allow      |
| *           | All traffic | All      | All        | 0.0.0.0/0   | Deny       |

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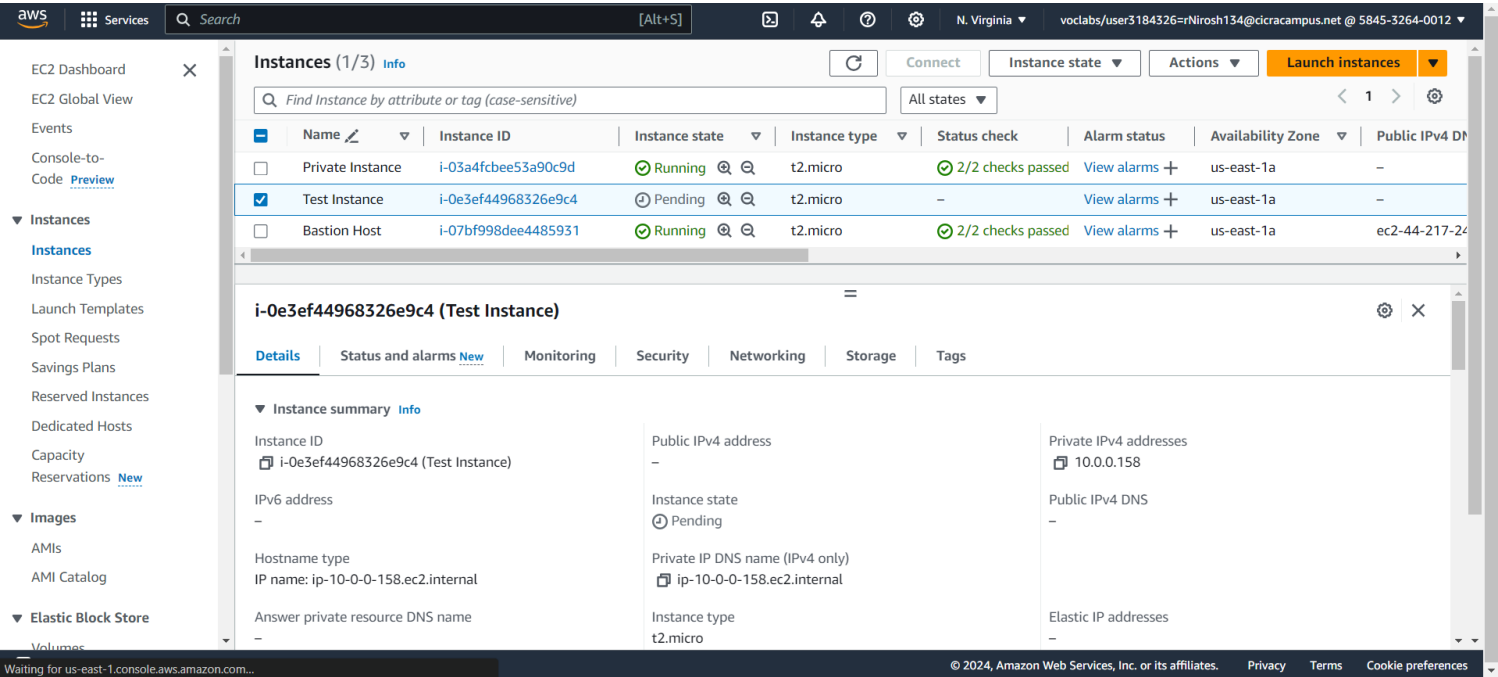
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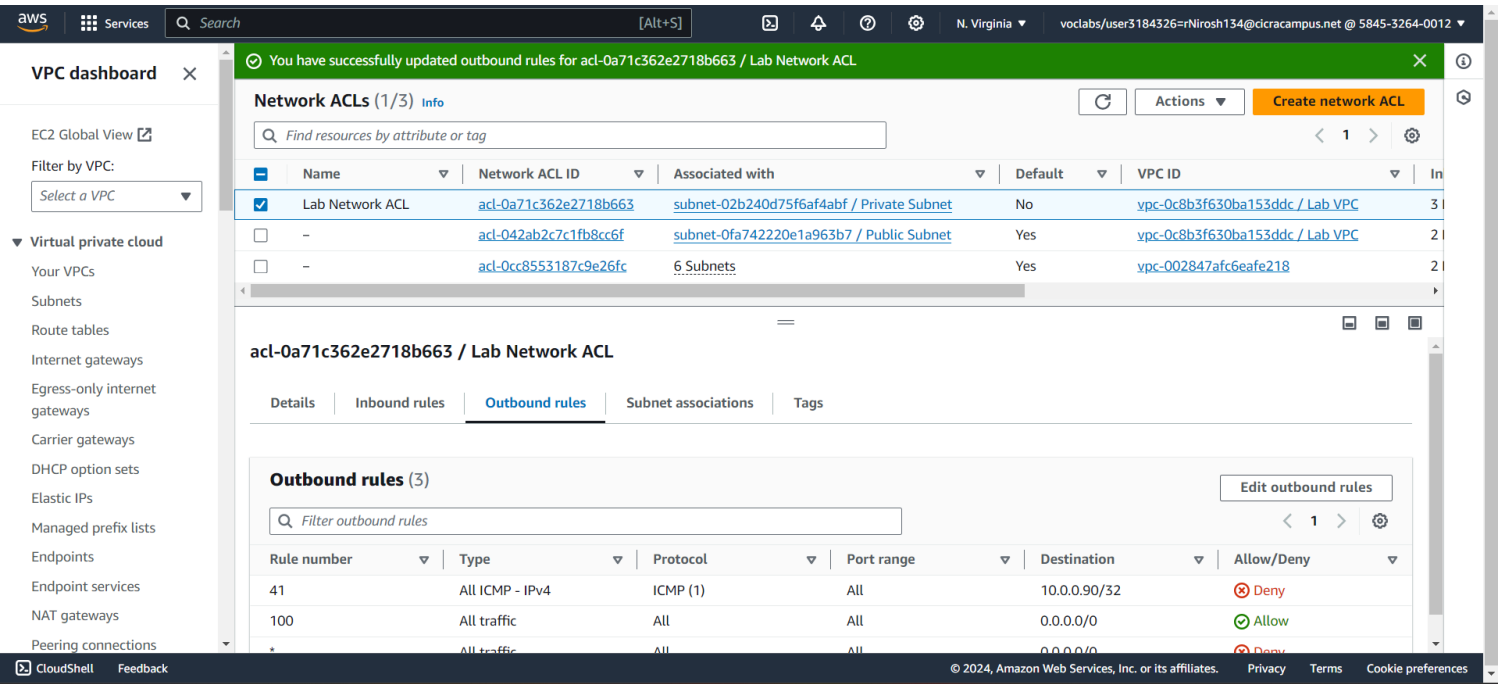
Cookie preferences

Task 11: Testing your custom network ACL

First we will create an instance in the EC2 called Test Instance in the public subnet.



Then we will test the connection between the test instance and the private instance by pinging a message to the test instance and see if it replies.



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**You have successfully updated outbound rules for acl-0a71c362e2718b663 / Lab Network ACL**

**Network ACLs (1/3)**

Find resources by attribute or tag

| Name                                                | Network ACL ID        | Associated with                           | Default | VPC ID                          |
|-----------------------------------------------------|-----------------------|-------------------------------------------|---------|---------------------------------|
| <input checked="" type="checkbox"/> Lab Network ACL | acl-0a71c362e2718b663 | subnet-02b240d75f6af4abf / Private Subnet | No      | vpc-0c8b3f630ba153ddc / Lab VPC |
| <input type="checkbox"/> -                          | acl-042ab2c7c1fb8cc6f | subnet-0fa742220e1a963b7 / Public Subnet  | Yes     | vpc-0c8b3f630ba153ddc / Lab VPC |
| <input type="checkbox"/> -                          | acl-0cc8553187c9e26fc | 6 Subnets                                 | Yes     | vpc-002847afc6eafe218           |

**acl-0a71c362e2718b663 / Lab Network ACL**

Details **Inbound rules** Outbound rules Subnet associations Tags

**Inbound rules (3)**

Filter inbound rules

| Rule number | Type            | Protocol | Port range | Source       | Allow/Deny |
|-------------|-----------------|----------|------------|--------------|------------|
| 41          | All ICMP - IPv4 | ICMP (1) | All        | 10.0.0.90/32 | Deny       |
| 100         | All traffic     | All      | All        | 0.0.0.0/0    | Allow      |
| *           | All traffic     | All      | All        | 0.0.0.0/0    | Deny       |

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Then we will set the ACL settings to stop the private instance from replying to the test instance, after applying this change, the ping message from the private instance won't get through to the Test instance anymore.

```
ec2-user@ip-10-0-1-160:~
64 bytes from 10.0.0.90: icmp_seq=142 ttl=127 time=0.459 ms
64 bytes from 10.0.0.90: icmp_seq=143 ttl=127 time=0.427 ms
64 bytes from 10.0.0.90: icmp_seq=144 ttl=127 time=0.415 ms
64 bytes from 10.0.0.90: icmp_seq=145 ttl=127 time=0.469 ms
64 bytes from 10.0.0.90: icmp_seq=146 ttl=127 time=0.438 ms
64 bytes from 10.0.0.90: icmp_seq=147 ttl=127 time=0.456 ms
64 bytes from 10.0.0.90: icmp_seq=148 ttl=127 time=0.441 ms
64 bytes from 10.0.0.90: icmp_seq=149 ttl=127 time=0.417 ms
64 bytes from 10.0.0.90: icmp_seq=150 ttl=127 time=0.502 ms
64 bytes from 10.0.0.90: icmp_seq=151 ttl=127 time=1.37 ms
64 bytes from 10.0.0.90: icmp_seq=152 ttl=127 time=0.463 ms
64 bytes from 10.0.0.90: icmp_seq=153 ttl=127 time=0.445 ms
64 bytes from 10.0.0.90: icmp_seq=154 ttl=127 time=1.13 ms
64 bytes from 10.0.0.90: icmp_seq=155 ttl=127 time=0.490 ms
64 bytes from 10.0.0.90: icmp_seq=156 ttl=127 time=0.551 ms
64 bytes from 10.0.0.90: icmp_seq=157 ttl=127 time=0.633 ms
64 bytes from 10.0.0.90: icmp_seq=158 ttl=127 time=0.429 ms
64 bytes from 10.0.0.90: icmp_seq=159 ttl=127 time=11.2 ms
64 bytes from 10.0.0.90: icmp_seq=160 ttl=127 time=0.528 ms
64 bytes from 10.0.0.90: icmp_seq=161 ttl=127 time=0.485 ms
64 bytes from 10.0.0.90: icmp_seq=162 ttl=127 time=0.460 ms
64 bytes from 10.0.0.90: icmp_seq=163 ttl=127 time=0.441 ms
64 bytes from 10.0.0.90: icmp_seq=164 ttl=127 time=0.406 ms
64 bytes from 10.0.0.90: icmp_seq=165 ttl=127 time=0.450 ms
64 bytes from 10.0.0.90: icmp_seq=166 ttl=127 time=0.416 ms
64 bytes from 10.0.0.90: icmp_seq=167 ttl=127 time=0.415 ms
64 bytes from 10.0.0.90: icmp_seq=168 ttl=127 time=0.460 ms
64 bytes from 10.0.0.90: icmp_seq=169 ttl=127 time=0.512 ms
64 bytes from 10.0.0.90: icmp_seq=170 ttl=127 time=0.466 ms
64 bytes from 10.0.0.90: icmp_seq=171 ttl=127 time=0.513 ms
64 bytes from 10.0.0.90: icmp_seq=172 ttl=127 time=0.472 ms
64 bytes from 10.0.0.90: icmp_seq=173 ttl=127 time=0.432 ms
64 bytes from 10.0.0.90: icmp_seq=174 ttl=127 time=0.425 ms
64 bytes from 10.0.0.90: icmp_seq=175 ttl=127 time=0.458 ms
64 bytes from 10.0.0.90: icmp_seq=176 ttl=127 time=0.402 ms
64 bytes from 10.0.0.90: icmp_seq=177 ttl=127 time=0.477 ms
64 bytes from 10.0.0.90: icmp_seq=178 ttl=127 time=0.471 ms
64 bytes from 10.0.0.90: icmp_seq=179 ttl=127 time=0.448 ms
64 bytes from 10.0.0.90: icmp_seq=180 ttl=127 time=0.378 ms
64 bytes from 10.0.0.90: icmp_seq=181 ttl=127 time=0.489 ms
64 bytes from 10.0.0.90: icmp_seq=182 ttl=127 time=0.414 ms
64 bytes from 10.0.0.90: icmp_seq=183 ttl=127 time=0.423 ms
```



Grade:

Mt grade is 56/56

aws academy

Account

Dashboard

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Help

ACAv2EN... > Assignments

> Module 6 Challenge Lab - Creating a VPC Networking Environment for the Cafe

Module 6 Challenge Lab - Creating a VPC Networking Environment for the Cafe

Due

No Due Date

Points 100

Submitting an external tool

Submit

Details

AWS

Start Lab

End Lab

1:29

Instructions

Grades

Actions

Launch Terminal

Total score

56/56

[Task 1A] Public Subnet

[Task 1B] Lab VPC has IGW

[Task 1C] Public Subnet with Internet

[Task 2] Bastion Host exists

[Task 3] Bastion Host has public ip

[Task 5] Private Subnet Exists

Submission

May 5 at 3:20pm

Submission Details

Grade: 55.35714285714286

(100 pts possible)

Graded Anonymously: no

Comments:

No Comments